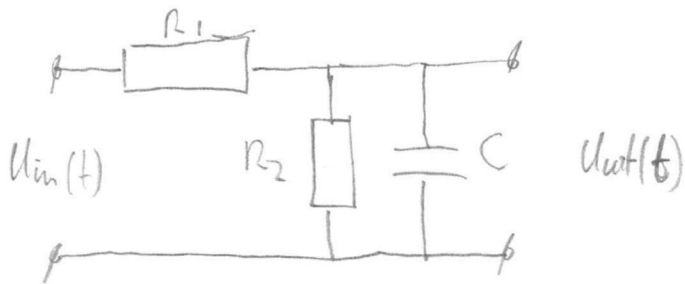


Üb. 12 gezeig 6a

HTS



$$U_{in}(t) = U_{R_1}(t) + U_{out}(t)$$

$$= R_1 \cdot i_{R_1}(t) + U_{out}(t) \quad (1)$$

* Nu $i_{R_1}(t)$ ausdrücken in R_2, C en U_{out} :

$$i_{R_1}(t) = i_{R_2}(t) + i_C(t) \quad (\text{strom in parallelschaltung})$$

$$= \frac{U_{R_2}(t)}{R_2} + C \cdot U'_C(t)$$

$$= \frac{1}{R_2} \cdot U_{out}(t) + C \cdot U'_{out}(t) \quad (U_{R_2} = U_C = U_{out}) \quad (2)$$

$$(1), (2) \Rightarrow U_{in}(t) = R_1 \left(\frac{1}{R_2} \cdot U_{out}(t) + C \cdot U'_{out}(t) \right) + U_{out}(t)$$

* Nu $U'_{out}(t) = s U_{out}(s)$ transformieren (Laplace-transformatie):

$$U_{in}(s) = R_1 \left(\frac{1}{R_2} \cdot U_{out}(s) + C s \cdot U_{out}(s) \right) + U_{out}(s)$$

$$R_2 \cdot U_{in}(s) = R_1 \cdot U_{out}(s) + R_1 \cdot R_2 \cdot C \cdot s \cdot U_{out}(s) + U_{out}(s) \cdot R_2$$

$$= \cancel{(R_1 + R_2 + R_1 R_2 C s)} \cdot U_{out}(s) \quad (R_1 + R_1 R_2 C s + R_2) \cdot U_{out}$$

$$\frac{U_{out}(s)}{U_{in}(s)} = \frac{R_2}{R_1 + R_1 R_2 C s + R_2}$$